Model-Free Multi-Agent Reinforcement Learning vs. Model Predictive Control for Heavy-Haul Train Platoons on Long Downhill Sections

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Abstract—Heavy-haul trains on long downhill sections (Shuozhou–Huanghua, 23.8 km, −10.9‰) rely on minute-scale cyclic air braking; when several trains form a virtually-coupled platoon, each must follow the cyclic-braking rhythm while keeping a safe inter-train gap and coupler force within limits. Model predictive control (MPC) is the industrial incumbent for this regime, but we argue it carries four structural limitations here: fragile finite-horizon/terminal feasibility, inability to reuse driving experience, dependence on an extremely precise multi-body model, and weak generalization across trains and lines. We demonstrate empirically that a fair multi-train MPC with the full coupler and gap model "sees" only a ≈ 3.2 s rolling window and cannot anticipate the minute-level brake–release rhythm: under a randomized-initialization protocol it stalls below the release limit for 71.6% of the run, exceeds the coupler limit (1,178–1,431 kN), and collapses into rear-end collision (330 collision steps, reward $-258{,}831 \pm 116{,}023$). A model-free hierarchical multi-agent reinforcement learning policy (M-MARL, rule-based cyclic braking with a hard coupler/ATP safety layer as the inner control layer + MAPPO cooperative thresholds as the outer layer) holds spacing and coupler force and never collides (reward +36,004, 0 collisions), and generalizes across randomized initial conditions, 5-train platoons, and a second 20 km profile (+36,120, 0 collisions). We map each MPC limitation to a concrete MARL advantage and a controlled head-to-head result.

Index Terms—Heavy-haul train, virtual coupling, model predictive control, multi-agent reinforcement learning, cyclic air braking, long downhill, coupler force.

I. Introduction

Heavy-haul trains operating on long downhill sections (e.g., the Shuozhou–Huanghua line, 23.8 km with sustained −10.9‰gradients) must regulate speed through cyclic air braking: braking whenever speed exceeds an upper limit and releasing when it falls below a lower limit. The minute-scale "brake–release" cycles, combined with multi-body coupler dynamics, produce large in-train longitudinal (coupler) forces that must stay below the safety limit (1000 kN normal / 2250 kN emergency). When several such trains operate as a virtually-coupled platoon to raise line capacity, each unit must additionally keep a safe inter-train spacing while following the cyclic-braking rhythm. This couples long-horizon, nonlinear, coupled, switched dynamics (air-brake wave propagation, brake–release hysteresis, varying consist and grade) with a safety-critical spacing constraint.

MPC is the incumbent, industrially applied method for train operation and, more recently, virtual coupling: Yang et al. (ITSC 2025) use single-train MPC on this same Shuozhou profile to minimize coupler impulse; Wang et al. (2026) apply Kalman-filtered MPC to cooperative heavy-haul virtual coupling (inter-train gap error ≤ 20 m); Song et al. (2026) derive a Tube-MPC for heavy-haul virtually-coupled train sets. It is therefore the natural baseline and the strongest reference point.

Yet we argue that MPC carries four structural limitations in exactly this regime, and that a model-free, global-horizon cooperative policy (multi-agent reinforcement learning, MARL) is the better fit:

1) Finite-horizon feasibility is fragile; terminal constraints are hard to design. MPC solves a local receding-horizon optimization. For nonlinear coupled systems a control-invariant terminal set must be imposed for recursive feasibility, but designing one is notoriously difficult [1]; when the horizon is too short to reach it the QP/NLP is infeasible, and lengthening the horizon raises computation and, under non-convexity, degrades solver reliability [2]. In our cyclic-braking platoon the relevant dynamics time-scale is minutes, far beyond a real-time-feasible horizon.
2) It cannot reuse driving expertise / past experience. A vanilla MPC is an optimization over a model each step; it has no mechanism to ingest driver braking practice or to improve from past runs. Learning must be bolted on (LMPC [3]).
3) It requires an extremely precise model. SLSQP/MPC relies on an accurate forward model of coupler forces, brake-wave propagation and brake–release hysteresis — exactly the components that are hardest to model precisely. With model mismatch MPC has no buffer (our simplified MPC without a coupler model overflowed the coupler limit).
4) It does not generalize across trains and lines. Each consist differs (mass, length, brake characteristics),

and MPC must be re-tuned and re-identified per train and per section; it optimizes locally and does not transfer learned structure across operating conditions.

Our thesis. Model-free MARL is not a "less rigorous" alternative to MPC; it is structurally better suited here: it (i) learns a global long-horizon driving rhythm instead of a shrinking local window, (ii) can be seeded/regularized with driver rules (rule-based cyclic braking as a safety-embedded inner layer), (iii) is model-free, so it does not need a perfect coupler/braking model, and (iv) generalizes across randomized initial conditions, larger platoons, and unseen profiles.

We demonstrate this on the real Shuozhou profile with 10,000-tonne trains and a multi-train platoon. Under the same randomized-initialization protocol our hierarchical MARL policy (M-MARL) holds spacing and coupler force and never collides (reward +36,004; 0 collisions), whereas a fair multi-train MPC with the full model, coupler constraint, and gap constraints collapses: speed stalls below the release limit for 71.6% of run time, coupler force exceeds the limit (1,178–1,431 kN), and under randomized initial conditions the spacing goes negative (rear-end collision, 330 collision steps; reward $-258{,}831 \pm 116{,}023$). The collapse maps directly onto MPC's short-rolling-window blind spot: its ≈ 3.2 s horizon cannot foresee the minute-level brake–release rhythm.

Contributions. (C1) We formalize the four structural limitations of MPC for long-downhill cyclic-braking platoon control and map each to a MARL advantage; (C2) we provide a controlled head-to-head (same environment, same safety layer, randomized protocol) showing MPC's receding-horizon blind spot empirically collapses the run while MARL transfers; (C3) we show MARL's generalization (50 random initial conditions, 5-train platoons, a second 20-km profile) that model-driven MPC cannot offer for free.

II. Related Work

MPC is the industrial incumbent for train control and virtual coupling. Yang et al. (ITSC 2025) use single-train multi-body MPC with cyclic air braking and coupler-impulse minimization on the Shuozhou profile; Wang et al. (2026) extend improved MPC with Kalman filtering for heavy-haul virtual coupling (gap error ≤ 20 m); Song et al. (2026) construct a Tube-MPC for heavy-haul virtually-coupled train sets with explicit terminal controllers and terminal constraint sets to certify stability; Lu et al. (ICIRT 2025) use NMPC for multiple locomotives of a single train. These confirm MPC is both the practical standard and an active research object.

MPC's known theoretical fragility (Lim 1). Stability and recursive feasibility of MPC rest on a control-invariant terminal set; designing one is "particularly difficult, especially in dynamic environments" [1], and short horizons can make the NLP infeasible while longer horizons raise cost and, under non-convexity, hurt solver reliability [2]. Surveys of virtual-coupling control classify MPC's main limitation as its finite-horizon/optimization burden relative to real-time coupling requirements [4].

Learning is being grafted onto MPC (Lim 2 & 3). LMPC incorporates data-driven learning of unmodeled dynamics into the optimization [3], acknowledging that plain MPC cannot learn from experience and is sensitive to model error. Robust/learning variants are themselves active research — and the very need to add learning is evidence that the plain method is being patched rather than intrinsically capable.

Model-free / RL turn (our approach). Roadmaps for RL-based virtual coupling [5] and coordinated-operation MARL [6] show growing evidence that model-free methods handle the constraint-satisfaction complexity that model-driven local optimization finds hard. Our hierarchical design (rule-based cyclic braking as a safety-embedded inner/control layer + MARL for the cooperative outer layer) is the model-free alternative that directly targets the four MPC limitations.

III. Problem Statement and System Model

Heavy-haul platoon of N trains, each a multi-body unit (1 locomotive + $\approx$ 100 wagons $\approx$ 10,000 t) on a long downhill grade profile. Longitudinal dynamics per vehicle: traction − braking − Davis resistance ± grade − coupler forces; air braking propagates along the consist at finite wave speed (≈ 200 m/s $\rightarrow$ seconds of delay across a ≈ 4 km train). Coupler forces arise from a mass-spring-damper model with a gap dead-zone. Control objective: keep each train speed in $[v_{\min}, v_{\max}]$, keep inter-train gap $\geq$ safety threshold, and keep coupler force ≤ 1000 kN, over a crossed-minutes horizon. A short linear MPC cannot see this horizon or model the switched coupler/braking force exactly — the crux of the comparison.

IV. Proposed Method

We retain the hierarchical structure but re-frame it explicitly as the model-free answer to the four MPC limitations.

- Inner/control layer (model-based-rule, safety-embedded): rule-based cyclic braking (brake $> v_{high}$, release $< v_{low}$) plus a hard coupler-force/ATP safety layer. This encodes driver expertise (addresses Lim 2 by soft-seeding) and does not need an explicit predicted model (addresses Lim 3).
- Outer/cooperative layer (learned): MAPPO, one agent per train, observes its own state plus the leader's, and outputs a cooperative threshold offset — i.e., it learns the global, long-horizon cooperative rhythm that MPC's local window cannot represent (addresses Lim 1). The policy generalizes across initial conditions, platoons and sections (addresses Lim 4).

Design mapping (Table I).

TABLE I
MPC limitations → MARL advantages.

MPC limitation	MPC failure mode (evidence)	MARL advantage (this work)
1. Finite horizon / terminal feasibility	≈ 3.2 s window; can't see minute-level cyclic rhythm → 70.8% below release floor, gap −9 m, 330 collision steps	learns global long-horizon rhythm; holds gap & coupler force
2. No driving-experience reuse	one-step model optimization, no past-data intake; learning is an add-on (LMPC)	rule-based cyclic braking seeded inside the policy
3. Requires precise model	simplified MPC (no coupler model) → coupler force 1,431 kN overflow	model-free; no explicit coupler/braking predictor
4. Weak generalization	per-train / per-line re-identification	generalizes across 50 random init conditions, 5 platoons, second 20-km profile

V. Experiments

Protocol. Same environment (Shuozhou main profile, 3000 steps), same SafetyLayer, randomized initial speeds 55–75 km/h, 5 seeds. Fair MPC = full multi-body model + coupler $\leq$ 1000 kN + gap constraint, block-horizon $B = 4 \times 8 \approx 3.2$ s, SLSQP.

Main comparison (fixed init, deterministic): see Table II.

MPC under randomized initial conditions (5 seeds): reward $-258{,}831 \pm 116{,}023$, coupler $1{,}431 \pm 207$ kN (>limit), gap min -9 ± 94 m, 329.6 ± 384.2 collision steps, only 4.2 cyclic-braking cycles, and 56.5 ms/step solver time (vs. μs–ms for the learned policy).

Mapping to the four MPC limitations (core evidence):

- Lim 1 (receding-horizon blinkering): the ≈ 3.2 s window stalls the train (71.6%–70.8% below release floor) and collapses spacing (−9 m, 330 collisions). MARL's global policy holds 0 collisions.
- Lim 2/3 (model & experience): the fair MPC already has the full coupler model yet still overflows the coupler limit (1,178–1,431 kN); a simplified MPC (no coupler model) overflows more. MARL needs no coupler predictor and is seeded with rule-based braking, staying at 659 kN.
- Lim 4 (generalization): M-MARL keeps +36,120 / 0 collisions on an out-of-domain 20 km profile and scales to 5-train platoons; MPC is a per-run local optimum with no transfer.

Pressure / scenario ablations (supporting): the cooperative benefit grows monotonically as the target gap shrinks (+26,213 → +47,502), consistent with the spacing-limited regime being exactly where the model-free cooperative layer matters.

VI. Conclusion

We re-frame heavy-haul cyclic-braking platoon control as a regime in which MPC's four structural limitations — finite-horizon/terminal feasibility, inability to reuse driving experience, sensitivity to model error, and weak generalization — are intrinsic properties rather than tuning issues, and where model-free hierarchical MARL addresses each. Controlled experiments on the real Shuozhou profile show the fair MPC collapses under the minute-scale cyclic-braking rhythm (stall, coupler-force overflow, rear-end collision) while MARL holds spacing and coupler force and generalizes. Limitations: MARL does not provide the formal recursive-stability guarantee that tube/MPC methods can certify; for the nominal short-horizon case MPC remains competitive; deployment safety would require formal verification/ATP integration; and the RL statistical outlier (1/3 seeds of an earlier model) must be reported honestly.

References

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TABLE II
Main comparison, Shuozhou main profile (3000 steps). M-MARL reports 5-seed mean; others deterministic single-seed.

Method	reward	stall%	coupler (kN)	gap min (m)	collision steps
M-MARL (ours, 5-seed)	+36,004	1.3	659	92	0
No-C (independent, δ=0)	+9,791	~ 1	543	—	69
M-DQN-fair	−15,215	1.8	463	−116	157
M-MPC-fair	−115,622	71.6	1178	105	0
Coast (reference)	+33,871	0	667	183	0